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**YISHUN JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2018**

BIOLOGY**9744/02****HIGHER 2****29 AUGUST 2018****Paper 2 Structured Questions****Wed 1400 - 1600****2 hours**

Candidates answer on the Question Paper.
No Additional Materials are required.

**READ THESE INSTRUCTIONS FIRST**

Write your name and CTG in the spaces at the top of this page and on all separate answer paper used.

Write in dark blue or black pen only.

You may use a soft pencil for any diagrams or graphs.

Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where Appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE	
Paper 1	/ 30
Paper 2	
Q1	/8
Q2	/9
Q3	/ 10
Q4	/7
Q5	/ 13
Q6	/ 7
Q7	/12
Q8	/10
Q9	/11
Q10	/13
TOTAL	/ 100

This document consists of **25** printed pages and **3** blank pages.

Section A

Answer **all** questions.

- 1** The cell theory was developed in the 1830s. At the same time, it was proposed that living things arose spontaneously from non-living materials. This theory of spontaneous generation was later disproven, but the cell theory has stood the test of time to become widely accepted in the scientific community today.

(a) Outline the cell theory.

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Fig. 1.1 is an electron micrograph of part of a eukaryotic cell.

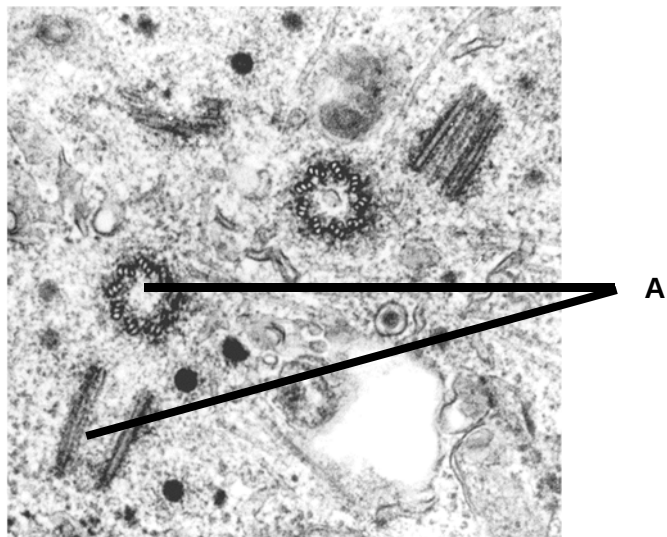


Fig 1.1

- (b)** State how it is possible to deduce that Fig. 1.1 is a transmission electron micrograph and **not** a scanning electron micrograph.

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.....[1]

(c) (i) Identify structure **A**.

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- (ii) Describe the structural features shown in Fig. 1.1 that identify **A**.

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- (iii) Cells such as that in Fig. 1.1 can divide by mitosis.

Explain the role of structure **A** in mitosis.

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[Total: 8]

- 2 Fig. 2.1 shows the structure of a triglyceride. Fig. 2.2 shows the structure of a glycolipid.

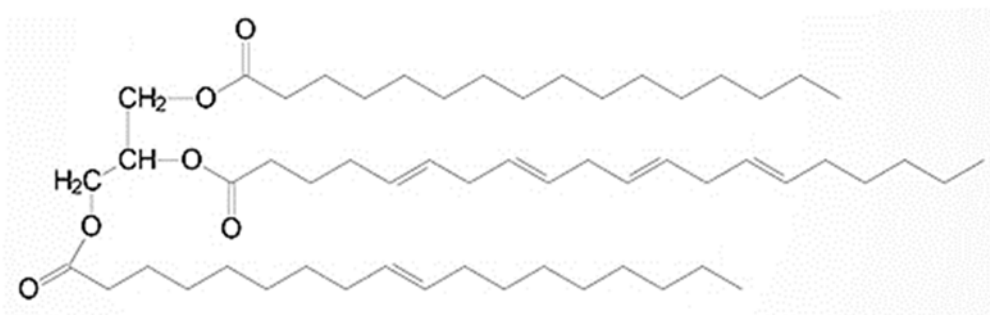


Fig. 2.1

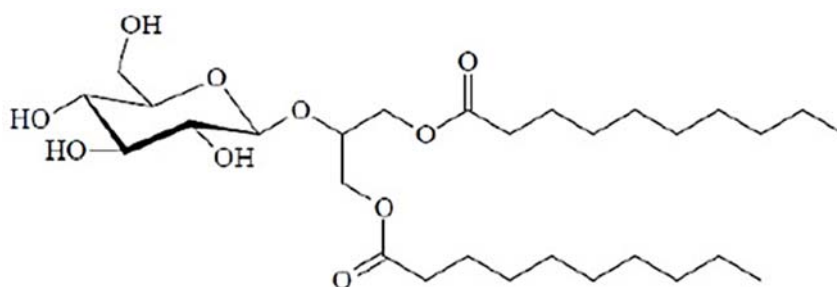


Fig. 2.1

- (a) Explain the difference in solubility between triglycerides and the products of their hydrolysis.

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- (b) Describe the roles of triglycerides and glycolipids in relation to the differences in their molecular structures.

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- (c) Suggest, in terms of membrane structure, how plants can be tolerant to freezing.

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[Total: 9]

- 3 High fructose corn syrup, made from maize, can be used as a replacement for sucrose to sweeten food and drink products.

Commercial production of high fructose corn syrup involves the enzyme glucose isomerase, extracted from bacteria.

Fig. 3.1 shows the results of an investigation into the effect of glucose concentration on the rate of reaction catalysed by glucose isomerase.

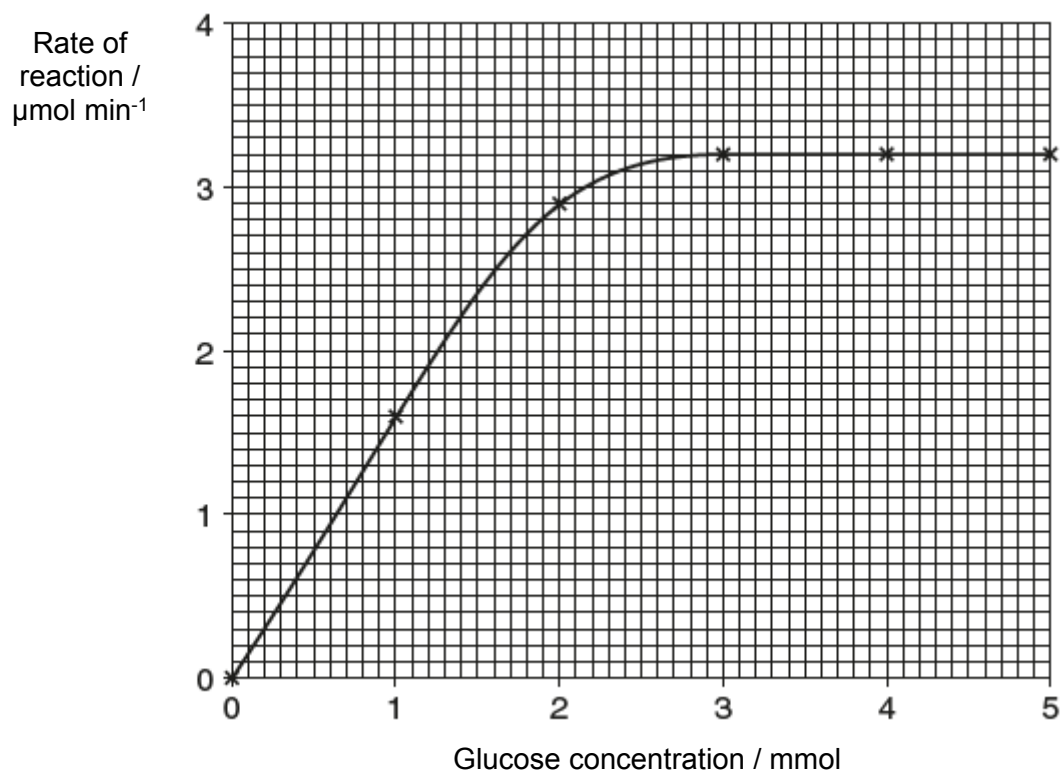


Fig. 3.1

(a) Using Fig. 3.1,

- (i) state the lowest substrate concentration that will give the maximum rate of reaction, V_{max} .

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- (ii) determine the Michaelis-Menten constant, K_m .

.....[1]

- (b) The glucose isomerase used in the production of high fructose corn syrup is extracted from a strain of a bacterium, *Thermus thermophilus*, which is found in hot springs. The enzyme has an optimum temperature of 95 °C.

Suggest **and** explain the advantages of using glucose isomerase from *T. thermophilus* to produce high fructose corn syrup, rather than glucose isomerase, that has an optimum temperature of 37 °C.

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The commercial production of high fructose corn syrup uses immobilised glucose isomerase.

Fig. 3.2 shows the effect of temperature on the activity of immobilised glucose isomerase compared to glucose isomerase free in solution.

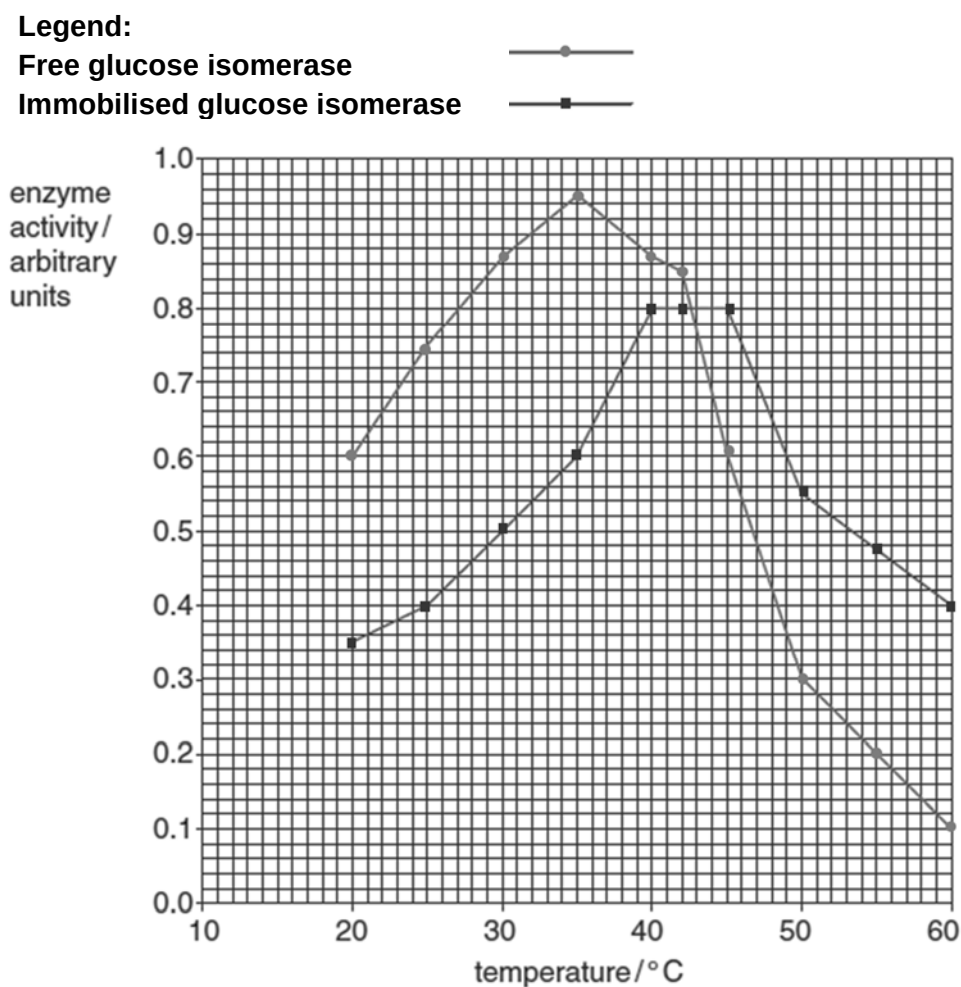


Fig. 3.2

(c) With reference to the Fig. 3.2,

- (i) describe and explain the effect of temperature on the activity of free glucose isomerase.

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- (ii) describe and explain the differences between the activity of free and immobilized glucose isomerases up to 40 °C.

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[Total: 10]

- 4 Fig. 4.1 shows information about the movement of chromatids in a cell that has just started metaphase of mitosis.

Legend

--- distance between chromatids

— distance between each chromatid and the pole to which it is moving

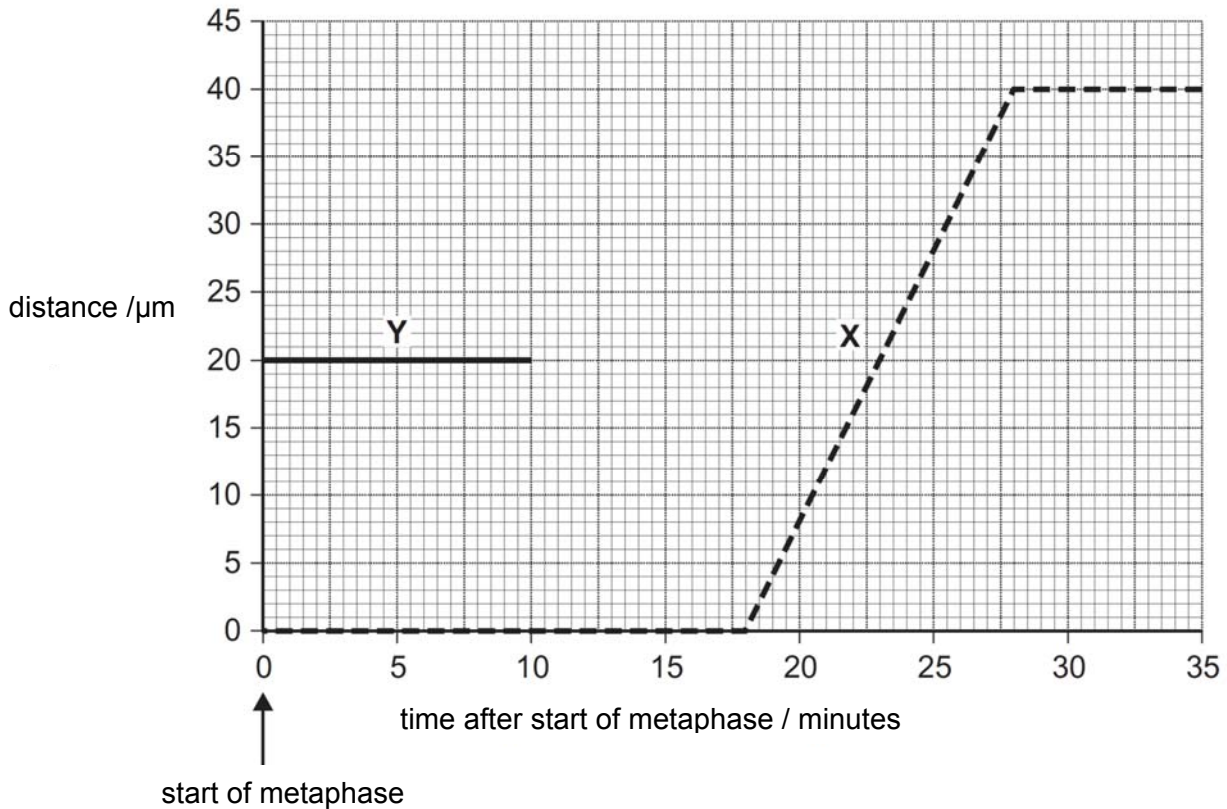


Fig. 4.1

With reference to Fig. 4.1,

- (a) (i) state the duration of metaphase in the cell.

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- (ii) complete line Y on the graph.

[1]

(iii) account for your answer in (a)(ii).

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The movement of chromatids is dependent on spindle fibres, which are made up of many tubulin subunits. Spindle fibres are lengthened at one end during mitosis by the polymerisation of tubulin subunits through GTP hydrolysis.

A drug, eribulin, is known to prevent the polymerisation of the tubulin subunits.

(b) Suggest and explain the effect of eribulin on the behaviour of chromosomes in mitosis.

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[Total: 7]

- 5 Erythropoietin, also known as EPO, is a large glycoprotein synthesised by specialised cells in the kidney. These cells are very sensitive to changes in oxygen concentration in the blood passing through the kidney and respond to a low oxygen concentration by increasing the synthesis of EPO.

EPO acts at the surface of particular target cells, such as cells in the bone marrow. These bone marrow cells are stimulated to produce red blood cells.

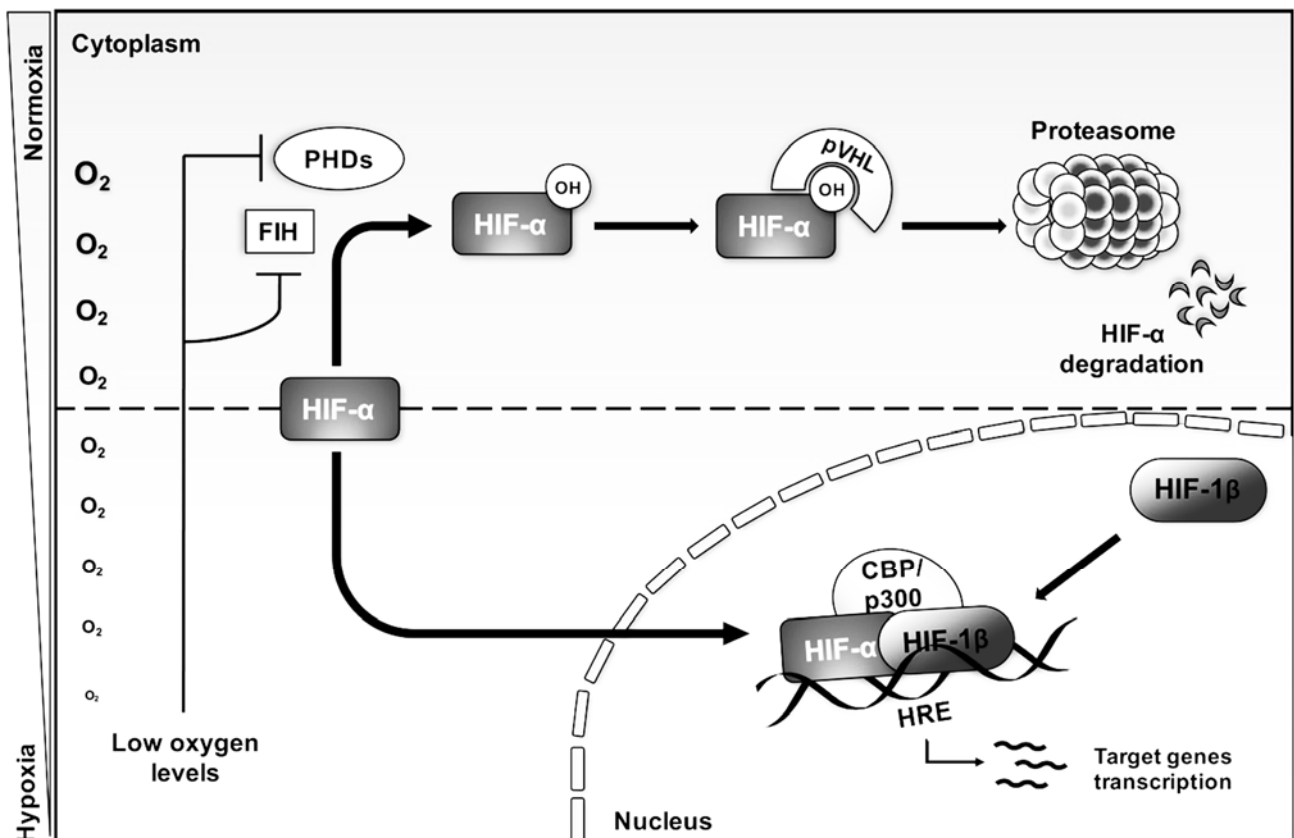
- (a) State precisely the location(s) where synthesis of the RNA(s) involved in EPO synthesis is / are located in the specialised cells in the kidney.

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[1]

Fig. 5.1 shows the effect of varying oxygen concentration on EPO gene expression.

Legend -

- CBP: CREB binding protein
- HIF: Hypoxia Inducible Factor
- HRE: Hypoxia response element



Biomedicines Review

Fig. 5.1

- (b) A low oxygen concentration (hypoxia) also leads to an increase in the quantity of mRNA in the specialised cells in the kidney.

With reference to Fig. 5.1 and your own knowledge of transcription, explain how the increase in the quantity of EPO mRNA is achieved.

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- (c) Protein synthesis accounts for a large proportion of the energy budget of a cell and therefore requires tight regulation.

Describe how translational control regulates gene expression.

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- (d) At high oxygen concentrations, HIF- α is degraded inside the proteasome. Suggest an advantage of degrading HIF- α at high oxygen concentrations.

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- (e) EPO cannot pass through the cell surface membrane to enter the bone marrow cell.

Suggest **one** reason why this is so.

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- (f) As part of an investigation into the body's response to EPO, a group of healthy young men were given injections of EPO every day for four weeks.

The haemoglobin (Hb) concentration for each subject was measured at the start of the investigation and then at intervals of one week for the next ten weeks. The first measurement was taken two weeks before the first EPO injection was given.

Fig. 5.2 shows the mean results for the subjects.

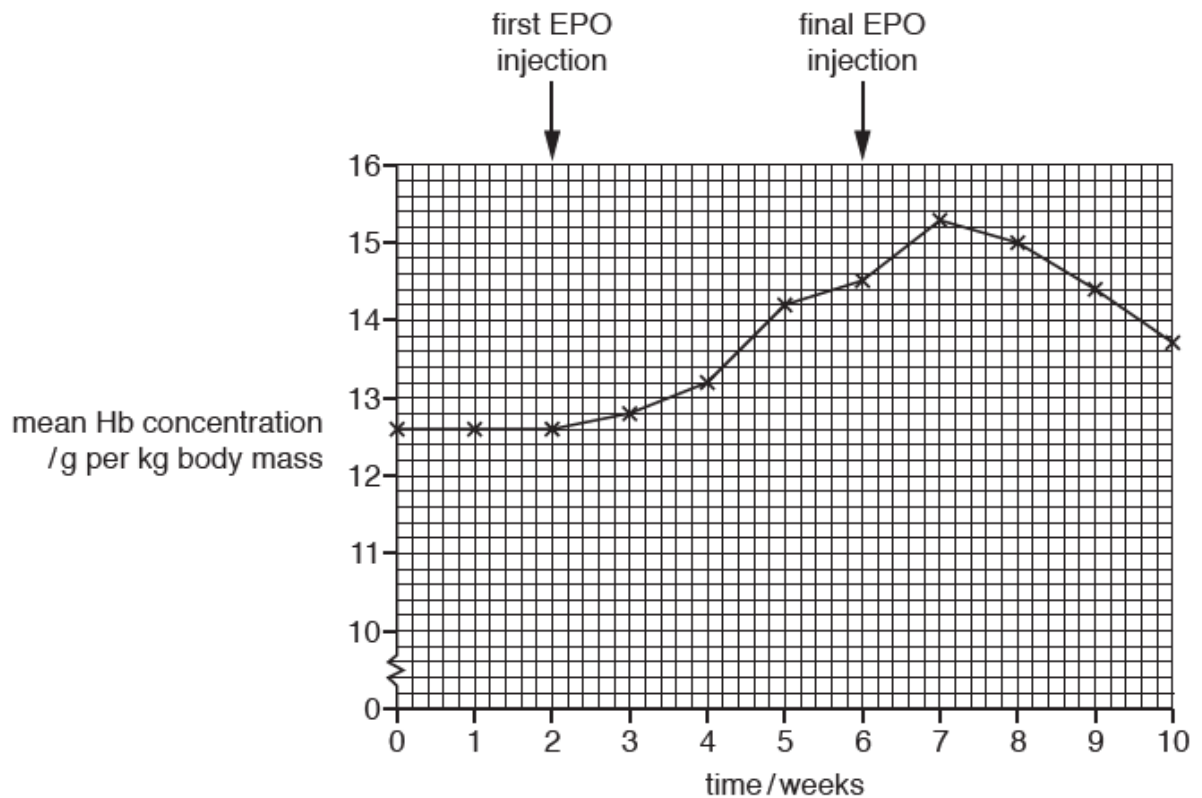


Fig. 5.2

Describe the results shown in Fig. 5.2 and suggest explanations for these results.

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[Total: 13]

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- 6 Normal muscle fibres can be generated if adult mesenchymal stem cells, without the genetic defect that causes Duchenne muscular dystrophy (DMD), is delivered into the patients' muscles. Mesenchymal stem cells can give rise to many types of cells including bone, cartilage, lung and muscle cells. They are known as adult stem cells.

(a) Describe **two** similarities between mesenchymal stem cells and cancer cells.

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[2]

Fig. 6.1 shows a possible approach for DMD gene therapy in humans to allow for the expression of full-length dystrophin protein. Dystrophin protein supports muscle fibre strength.

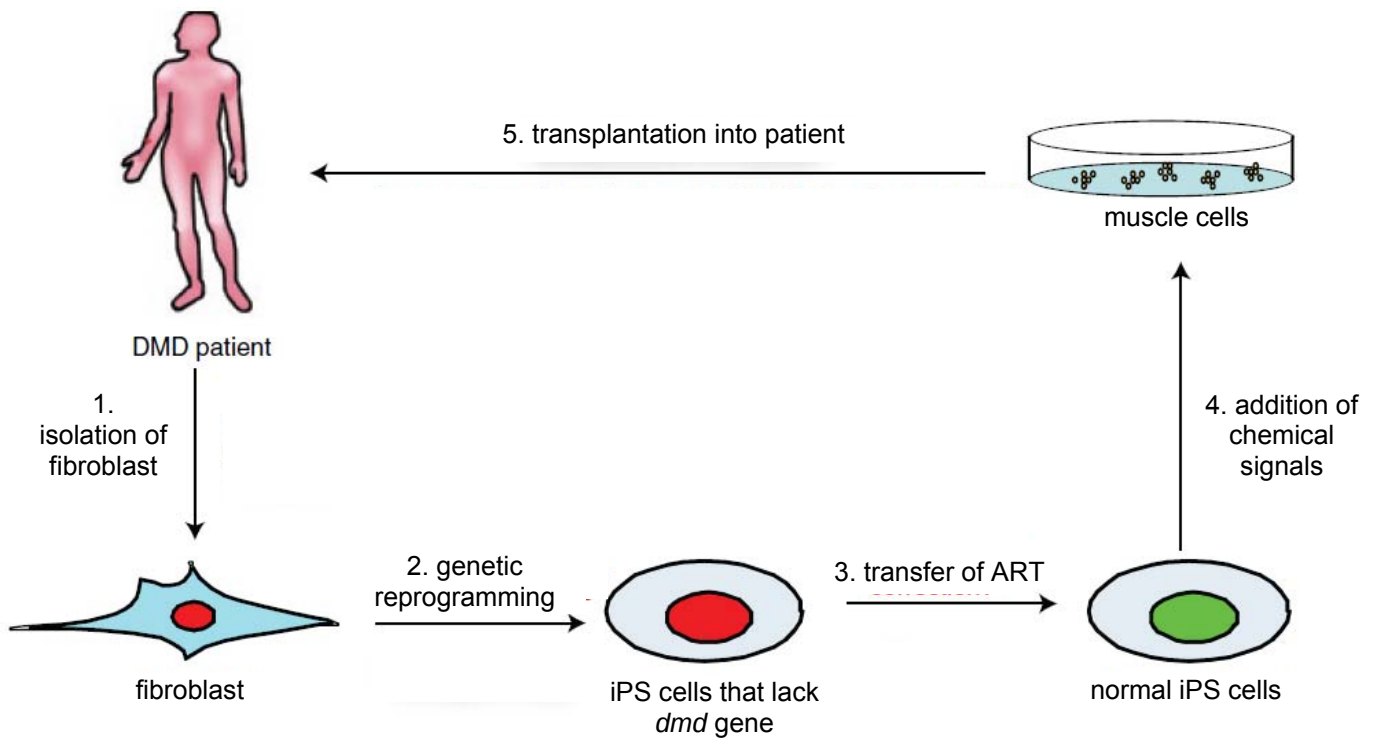


Fig. 6.1

- (b)** Explain why it is preferable to isolate fibroblast from the patient with DMD.

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- (c)** Upon transplantation into the patient, suggest disadvantages of this approach.

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[Total: 7]

- 7 Fig. 7.1 shows the process of binary fission in bacteria.

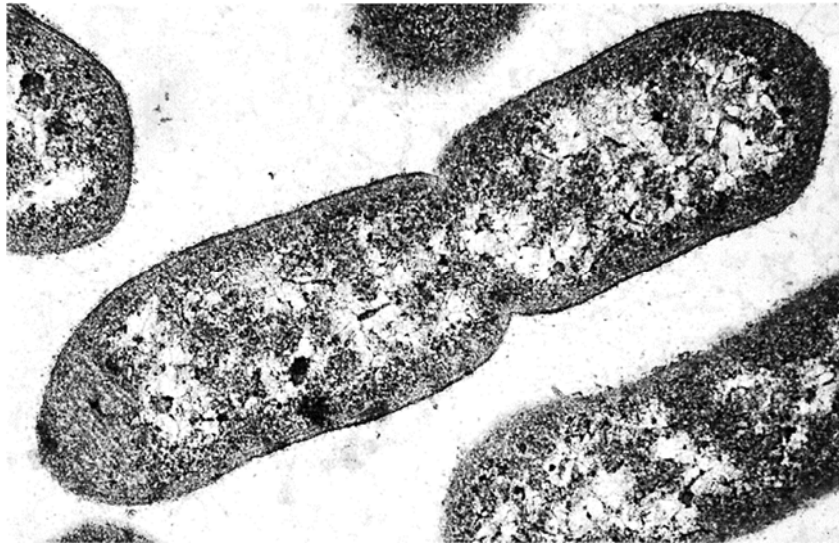


Fig. 7.1

- (a) (i)** Briefly describe the events occurring in binary fission.

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- (ii)** Contrast binary fission and the process of cell division in an animal cell.

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Recent findings suggest that the mucus lining the human digestive tract is loaded with bacteriophages. These bacteriophages contain proteins which bind to the carbohydrate residues of mucoproteins as shown in Fig. 7.2. It has been proposed that these bacteriophages can play an important role as part of the body's defense system.

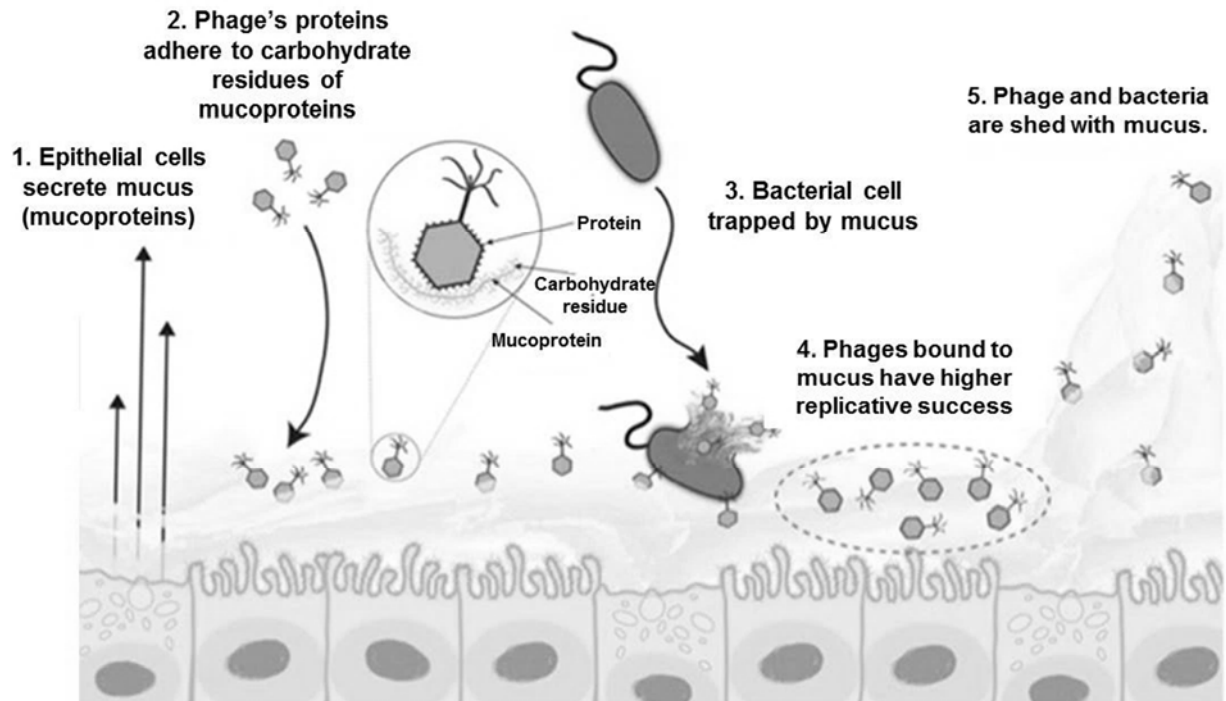


Fig. 7.2

(b) Describe the life cycle of the bacteriophages shown in Fig. 7.2.

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- (c) Many other viruses may also be found within the human body e.g. the Human Immunodeficiency Virus (HIV).

Compare the lysogenic life cycle of a temperate bacteriophage with the life cycle of a HIV.

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[Total: 12]

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- 8 In an experiment involving deer mouse, pure breeding pink-eyed mice with wild-type fur was crossed with pure breeding dark-eyed albino mice.

All the resulting progeny had wild-type fur and dark eyes. These F_1 mice were then crossed with pink-eyed albino mice. The results are shown in Table 8.1. It was difficult to distinguish between mice that are dark-eyed albino and pink-eyed albino, so these two phenotypes were counted together (F.H. Clark. 1936. *Journal of Hereditary* 27:259-260).

Table 8.1

Phenotype	Number of progeny
Dark-eyed, wild fur	0
Pink-eyed, wild fur	63
Dark-eyed, albino	71
Pink-eyed, albino	

A chi-squared (χ^2) test was carried out on the results of the second cross. The calculated X^2 value is 32.8. Table 8.2 shows part of the critical values of the chi-squared distribution.

Table 8.2

number of degrees of freedom (v)	distribution of X^2		
	probability		
	0.1	0.05	0.01
1	2.71	3.84	6.64
2	4.60	5.99	9.21
3	6.25	7.82	11.34
4	7.78	9.49	13.28

- (a) Using the χ^2 table, explain the conclusion drawn from the calculated χ^2 value obtained.

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- (b) (i) Using **D/d** to represent alleles for eye colour and **A/a** to represent alleles for fur coat colour, explain the observed results of the second cross in the deer mouse experiment using a genetic diagram.

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- (ii) How would you explain the experimental ratio?

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[Total: 10]

- 9 ATP and NAD both play important roles in respiration. Both compounds are nucleotides. Fig. 9.1 represents the molecular structures of ATP and NAD.

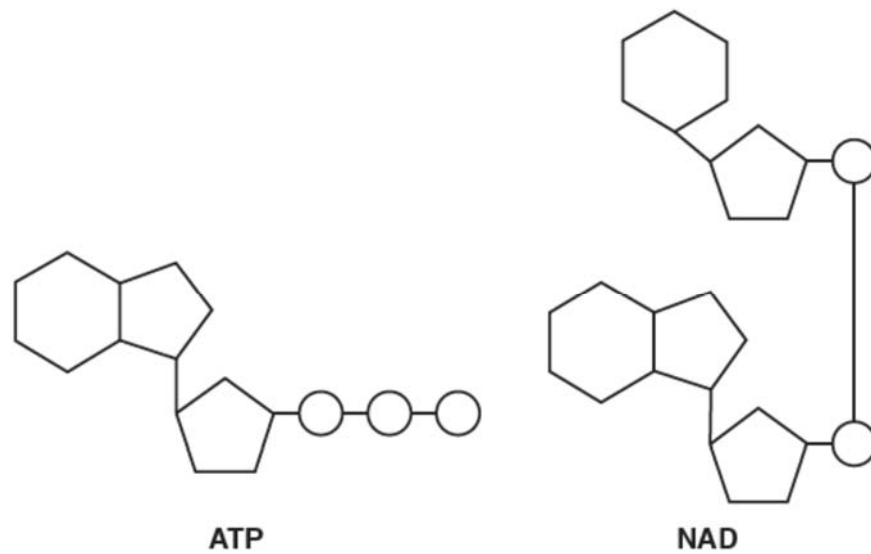


Fig. 9.1

- (a) Using Fig. 9.1, compare the structures of ATP and NAD.

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- (b) Name the type of chemical reaction by which ATP is made during the Krebs cycle.

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- (c) Outline the roles of NAD in the cytoplasm of a cell.

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- (d) An experiment was carried out to investigate the effect of epicatechin on mitochondrial respiration in mice. Epicatechin is a naturally occurring compound in cocoa beans and is present in chocolates.

Two groups of mice, group A and group B, were used in this experiment.

- Group A was given water containing epicatechin, twice a day for 15 days.
- Group B was given water without epicatechin, twice a day for 15 days.

After 15 days, the structure of mitochondria from striated muscle cells in both groups of mice was examined.

The surface area of the inner membrane of the mitochondria was divided by the surface area of the outer membrane to obtain a ratio for each mouse.

Table 9.1 shows the mean ratios for the two groups of mice.

Table 9.1

group	mean ratio
A	2.0 : 1
B	1.7 : 1

The mice in group **A** were able to exercise for a longer period of time than the mice in group **B**.

With reference to Table 9.1, explain why the mice in group **A** were able to exercise for a longer period than the mice in group **B**.

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[Total: 11]



- (b)** Streptomycin was the first antibiotic used to treat TB

During the first few years after the introduction of streptomycin treatment, an increasing number of *Mycobacterium tuberculosis* bacteria developed resistance to streptomycin.

- (i)** Outline how this happened.

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- (ii)** Suggest the mechanism of antibiotic resistance in these resistant strains of *M. tuberculosis* bacteria.

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- (c)** Vaccination has been used successfully to control the spread of many infectious diseases in the population.

Explain how vaccination works in terms of immunological memory and herd immunity.

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[Total: 13]

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